PRML COURSE PROJECT

Bone Age Prediction Project Report

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1. Abstract

This project predicts **bone age** from hand X-ray images using deep learning. Using the RSNA Bone Age dataset, we implemented multiple approaches including HOG features, baseline CNN, and MobileNet with fine-tuning.

We addressed the problem in **two ways**:

1. **Regression:** Predicting exact bone age (in months).
2. **Classification:** Categorizing bone age into developmental stages: Infant, Child, Pre-Adolescent, Adolescent, Adult-like.

Regression Results (MobileNet Fine-Tuned):

- MAE: 15.10 months
- RMSE: 20.20 months
- R^2: 0.765

Classification Results (MobileNet Fine-Tuned):

- Accuracy: 0.63

- Weighted F1-score: 0.63
- Pre-Adolescent stage predicted best; Adolescent stage worst

Grad-CAM visualizations indicate the model focuses on **clinically relevant regions**, such as wrist and finger joints.

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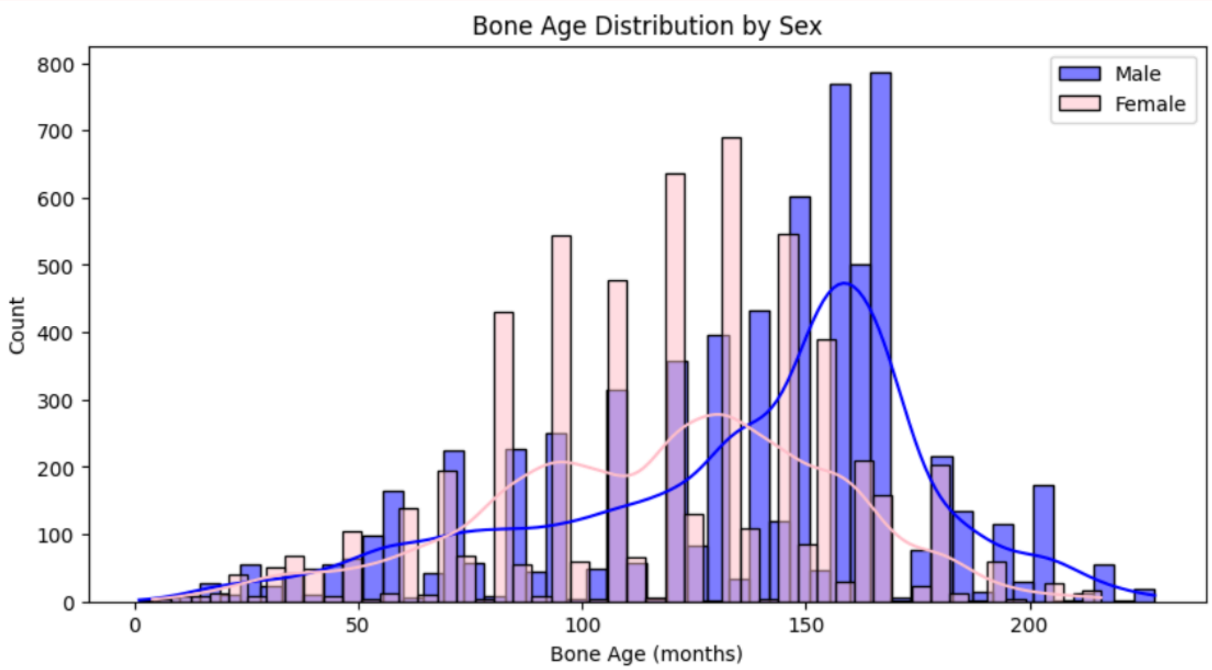
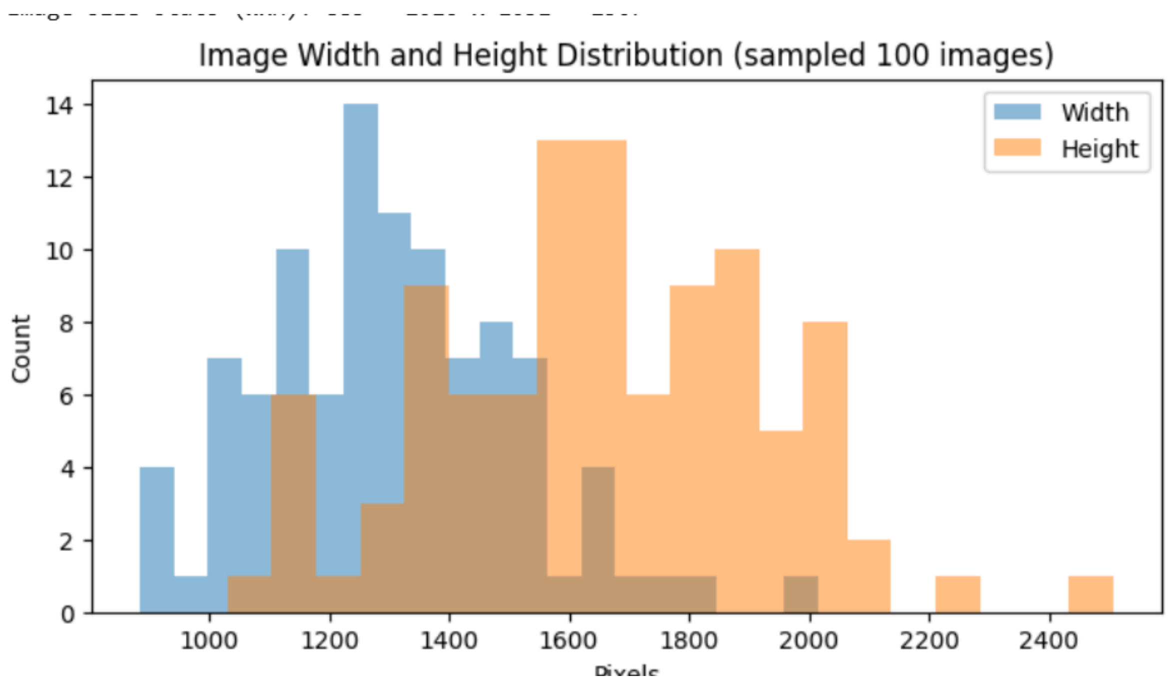
2. Methodology

2.1 Data Preprocessing

- Images resized to 224×224 pixels and normalized to [0,1]
- Grayscale images converted to 3 channels
- Train-validation split: 90%-10%
- GPU-enabled TensorFlow dataset pipeline for faster training
- Insights about dataset:





2.2 Regression Model

Stepwise Development and Metrics:

Step	MAE	RMSE	R^2
HOG + Random Forest	27.47	33.63	0.34
Baseline CNN	23.13	30.73	0.46
MobileNet Frozen	19.21	25.43	0.63

MobileNet Fine-Tuned	15.10	20.20	0.765

- **Model Design:** Pretrained MobileNet base + regression head, fine-tuned last 50 layers, optimizer: Adam, loss: MSE.
- **Observations:** Gradual improvement confirms the impact of pretrained CNN and fine-tuning.

2.3 Classification Model

- Target: Classify bone age into **5 developmental stages**.
- MobileNet-based classifier with transfer learning and early stopping (patience=3).
- **Training:** 25 epochs

Classification Metrics:

Class	Precision	Recall	F1-Score	Support
Infant	0.82	0.59	0.69	665
Child	0.54	0.46	0.50	68
Pre-Adolescent	0.58	0.92	0.71	142
Adolescent	0.83	0.29	0.43	17

Adult-like	0.48	0.64	0.55	370
Overall Accuracy			**0.63**	1262

- Confusion matrix shows minority classes (Adolescent, Adult-like) are harder to predict.
- Early stopping restored best weights and prevented overfitting.

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3. Results & Visualizations

Regression

- **Predicted vs Actual Scatter Plot:** Most points align close to the true line; few outliers at extreme ages.

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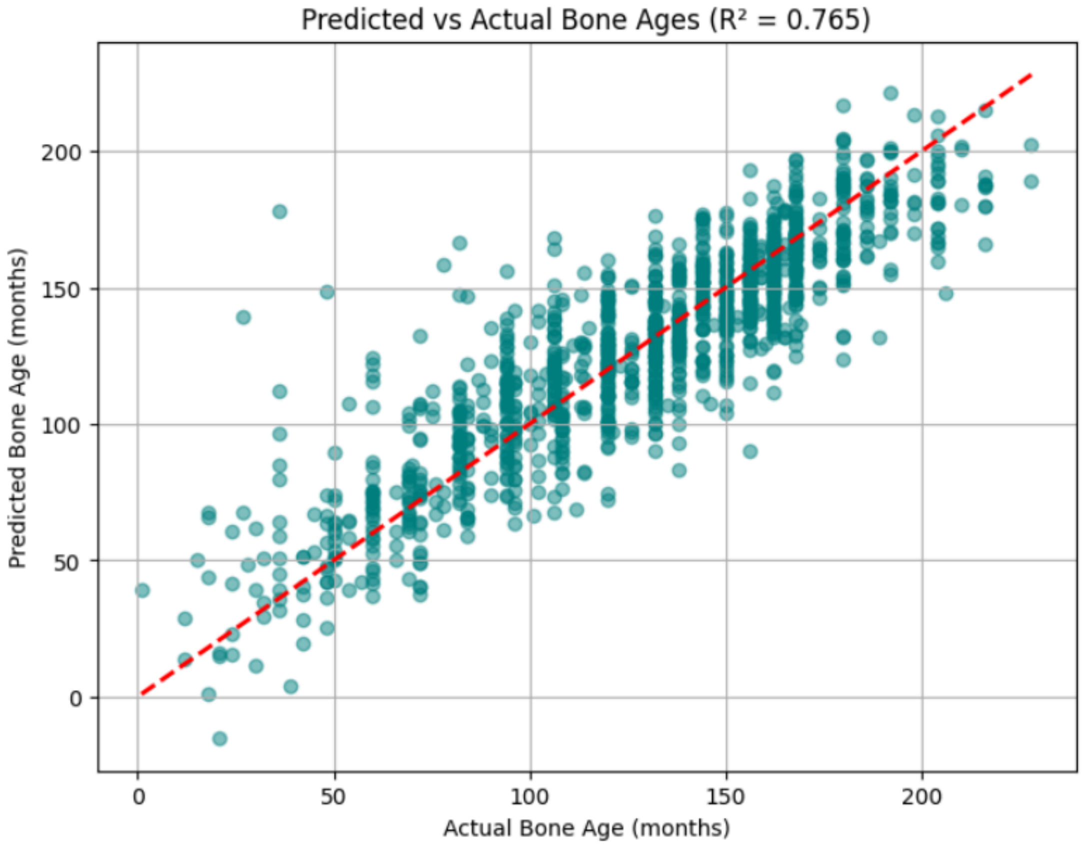
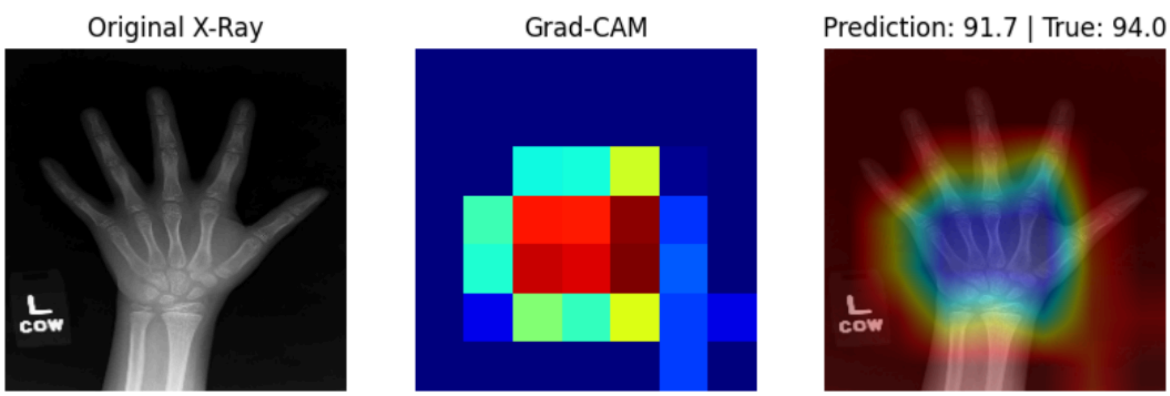
40/40 ━━━━━━━━━━━━━━━━━━━━ 11s 271ms/step
MAE  : 15.10
RMSE : 20.20
R²   : 0.765
```



- **Grad-CAM:** Highlights wrist and finger joints confirming model uses meaningful feature



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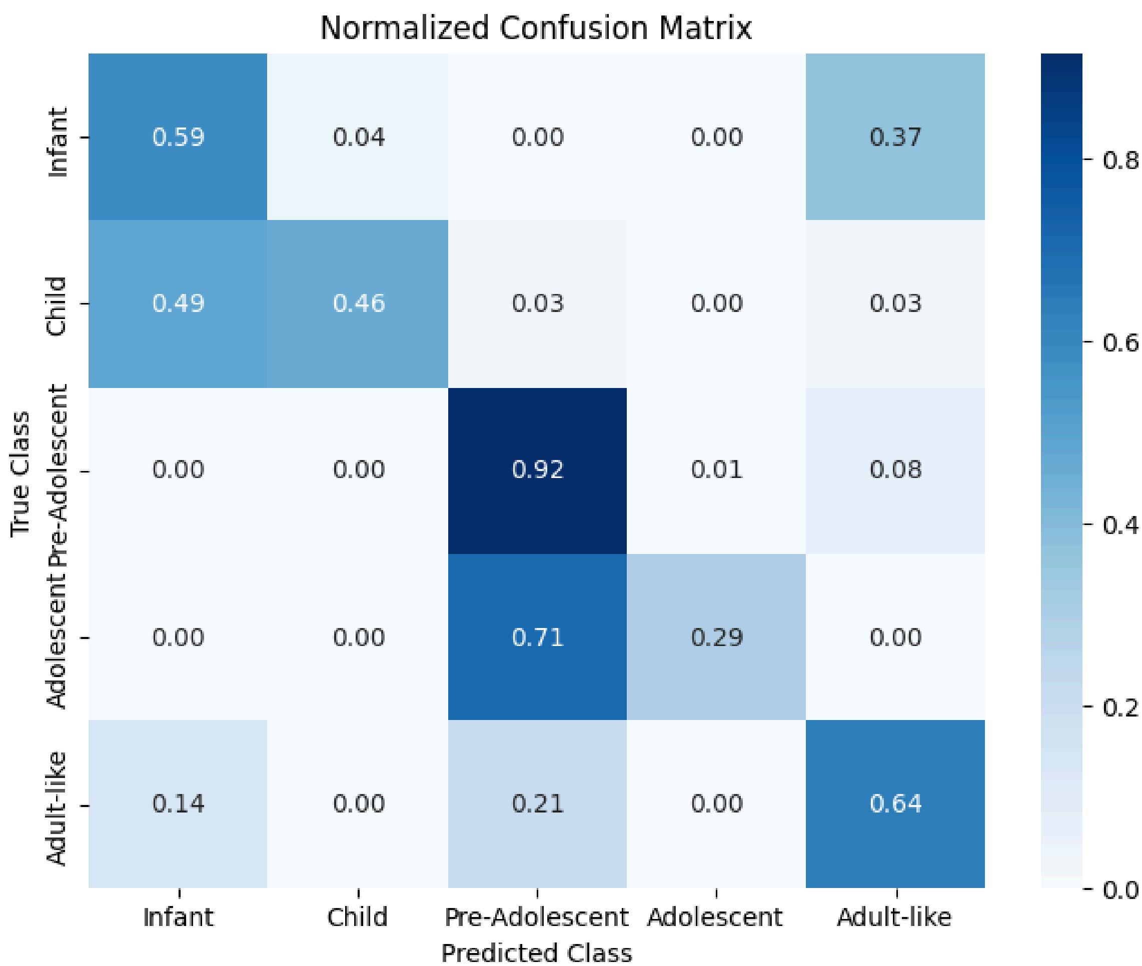
Male MAE: 15.25
Female MAE: 14.92
```

Classification

- **Confusion Matrix:** Shows which classes are confused.



- **Stage-wise Accuracy:** Pre-Adolescent stage has highest recall; Adolescent stage lowest.

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40/40 ━━━━━━━━━━━━━━━━━━━━ 11s 274ms/step
                precision    recall  f1-score   support

        Infant       0.82      0.59      0.69       665
         Child       0.54      0.46      0.50        68
Pre-Adolescent       0.58      0.92      0.71       142
    Adolescent       0.83      0.29      0.43        17
    Adult-like       0.48      0.64      0.55       370

      accuracy                           0.63      1262
     macro avg       0.65      0.58      0.57      1262
  weighted avg       0.68      0.63      0.63      1262
```

- **QWK Score:**

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40/40 ━━━━━━━━━━━━━━━━━━━━ 17s 331ms/step
QWK : 0.5325065532343021
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4. Discussion

- Regression is strong: R^2 of 0.765 (~76% variance explained).
- Classification is moderate: Accuracy 63%, F1-score 0.63
- Stepwise improvements show pretrained CNN + fine-tuning is crucial.
- Errors higher at extremes of bone age and minority classes.
- Future work:
    - Improve minority class prediction via augmentation or oversampling
    - Experiment with ensemble models
    - Combine regression and classification for hybrid approach

5. Conclusion

- Bone age prediction from X-ray images is feasible with deep learning.
- MobileNet fine-tuning significantly improves regression performance.
- Classification of developmental stages is reasonable but can be improved with more balanced data.

6. References

1. RSNA Bone Age Dataset: https://www.kaggle.com/datasets/kmader/rsna-bone-age
2. TensorFlow 2 Documentation: https://www.tensorflow.org
3. Chollet, F., *Deep Learning with Python*, 2nd Edition, 2021
4. Selvaraju, R. R. et al., Grad-CAM, 2017